REV	ECO#	DESCRIPTION	DATE
2		Move Silkscreen of R14 so it can be read. Swap RB4 and RC7 connections. Change @U4 to PIC16F1619-I/P	3-31-2015
2.2		Change @U4 to PIC16F18346-I/P	8-2-2018

USB INTERFACE (BUS POWERED)

DNP TP1 VIN J15 PJ-002BH-SMT DNP D3 US1M R59 2k 0603 VCMP R61 10k TP2 DNP GND +9V LM340MP-5.0 U5 DNP VIN VOUT GND C17 0.22uF C16 0.1uF 25V +5V

MCP1703-3302E/DB Q2 VIN VOUT GND +5V C10 0.22uF 16V C11 1uF 16V +3.3V C12 1uF 16V R26 1k R27 10k C13 0.1uF 25V C14 0.01uF 16V +3.3V D1 GREEN POWER_GOOD_PICKIT3

DNP R45 0R 0603 FDN340P Q5 D_VBUS +5V

USB MINI-B Female J2 VBUS D- D+ ID GND USB_D_N USB_D_P 1210 500mA PTC3 D_VBUS U3 VBUS054B-HS3-GS08 DNP

U16 MCP6561 VREF_2.5V VCMP +5V -A +A VDD OUTA VSS R60 10k GND

POWER HDR-2.54 Male 1x3 J12 +5V +3.3V DNP TP3 VDD DNP TP4 GND

HDR-2.54 Female 1x6 DNP J7 VIN +3.3V +5V GND

HDR-2.54 Female 1x8 DNP J33 RC2 RA4 CS SCK MISO MOSI +3.3V GND AN RST CS SCK MISO MOSI +3.3V GND PWM INT RX TX SCL SDA +5V GND mikroBUS J35 GND DNP J34 RC5 RA2 RX TX SCL SDA +5V HDR-2.54 Female 1x8

RB4 RB6 RC7 RC6 R55 0R MISO R53 0R SCK R57 0R MOSI R51 0R CS J39 HDR-2.54 Male 2x4 DNP

RX SCL SDA TX R52 0R RB5 R56 0R RB6 R58 0R RB4 R54 0R RB7 J40 HDR-2.54 Male 2x4 DNP

DNP J8 GND AIO2 AIO1 AIO0 RX TX WAKE_SW CMD/MLDP HDR-2.54 Male 1x8

LED/PIO1/SCK MLDP_EV/PIO2/CS WS/PIO3/MOSI PIO4/MISO CTS/PIO5 RC0 +3.3V RSVD2 RSVD1 RSVD0 PIO7 RTS/PIO6 SPI/PIO DNP J10 HDR-2.54 Male 1x8

GND J14 DNP HDR-2.54 Male 1x8 GND

HDR-2.54 Male 1x2 DNP J5 ICSP_VDD R46 0R VDD R35 470R C15 0.1uF 25V D2 GREEN GND

J9 VDD RA5 RA4 nMCLR RC5 RC4 RC3 RC6 RC7 RB7 HDR-2.54 Female 2x10

U4 110-91-320-41-001

J11 PGED PGEC RA2 RC0 RC1 RC2 RB4 RB5 RB6 HDR-2.54 Female 2x10 PGED PGEC RB5 GND

@U4 PIC16F18346-I/P VDD RA0/ICSPDAT RA1/ICSPCLK RA2 RA3/Vpp/MCLR RA4/OSC2 RA5/OSC1 RB4 RB5 RB6 RB7 VSS RC0 RC1 RC2 RC3 RC4 RC5 RC6 RC7

DNP P1 nMCLR ICSP_VDD PGED PGEC GND HDR-2.54 Male 1X6 STAGGERD

DNP J13 HDR-2.54 Male 2x4 RA5 PGEC RA2 RC5 R48 0R R49 0R R47 0R R50 0R

HDR-2.54 Male 1x2 DNP J4 RC4 R34 0R 0402 R31 470R R29 10k S1 TACT SPST GND VDD

DNP J3 RC0 R33 0R 0402 R32 470R POT1 10k HDR-2.54 Male 1x2 VDD GND

nMCLR R39 470R S2 TACT SPST GND VDD R38 10k

R37 10k R44 0R 0402 RC1 J6 HDR-2.54 Male 1x2 DNP S3 mTouch™ Button

R40 1k R41 1k R42 1k R43 1k D4 RED D5 RED D6 RED D7 RED GND

+3.3V U6 C18 4.7uF GND VDD GND GND GND GND AIO2 AIO1 AIO0 PIO7 RSVD0 RSVD1 RSVD2 AIO2 AIO1 AIO0 PIO7 RSVD RSVD RSVD UART_TX UART_RX CTS/PIO5 RTS/PIO6 WAKE_HW WAKE_SW CMD/MLDP LED/PIO1/SCK MLDP_EV/PIO2/CS WS/PIO3/MOSI PIO4/MISO SPI/PIO RX TX CTS/PIO5 RTS/PIO6 RC0 WAKE_SW CMD/MLDP LED/PIO1/SCK MLDP_EV/PIO2/CS WS/PIO3/MOSI PIO4/MISO SPI/PIO RN4020

X1 X2 X3 X4 X5

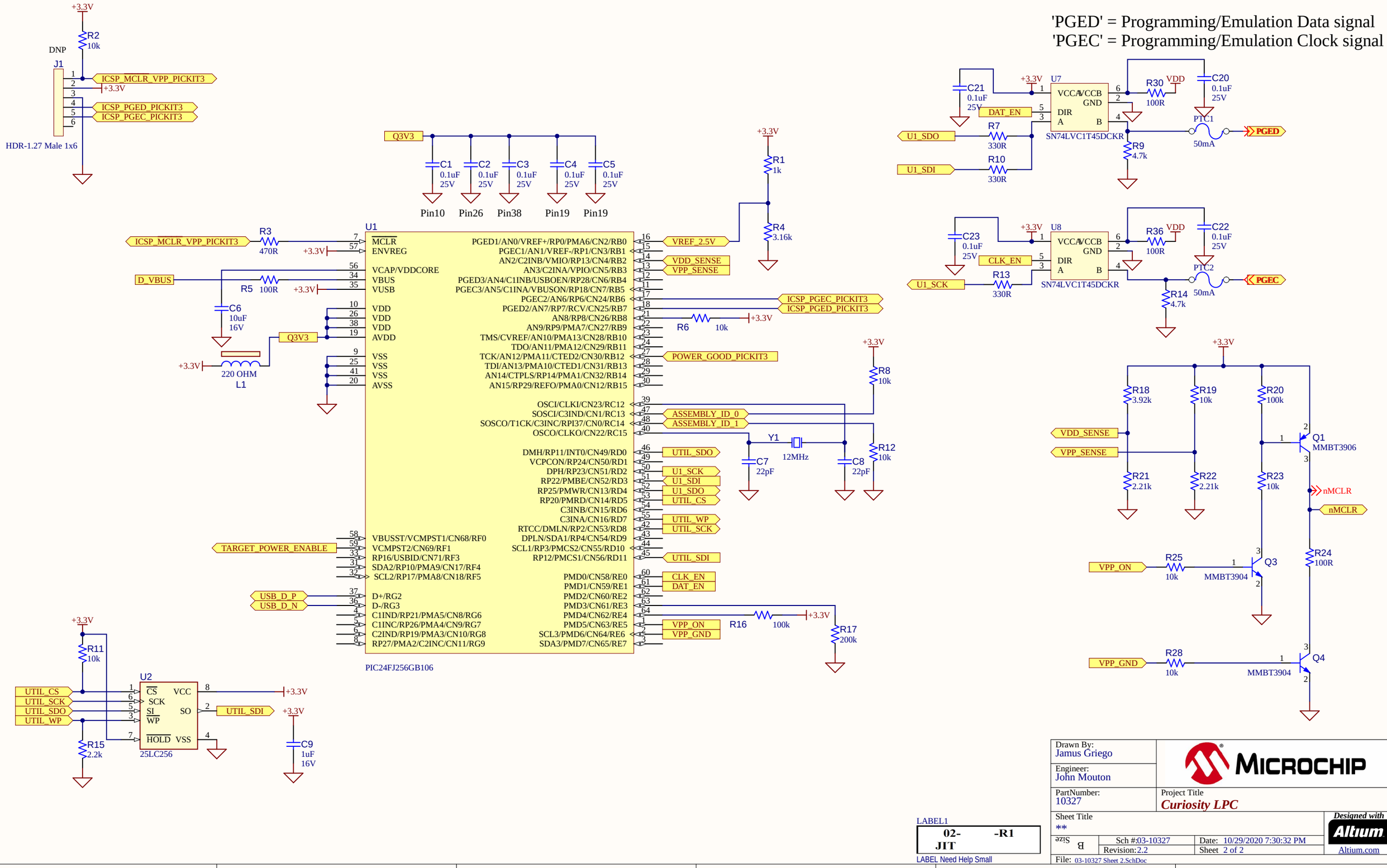
SCR1 Phillips Screw 1/4" SCR2 Phillips Screw 1/4" SCR3 Phillips Screw 1/4" SCR4 Phillips Screw 1/4"

STANDOFF1 STANDOFF2 STANDOFF3 STANDOFF4

1/4" Standoff Nylon 1/4" Standoff Nylon 1/4" Standoff Nylon 1/4" Standoff Nylon

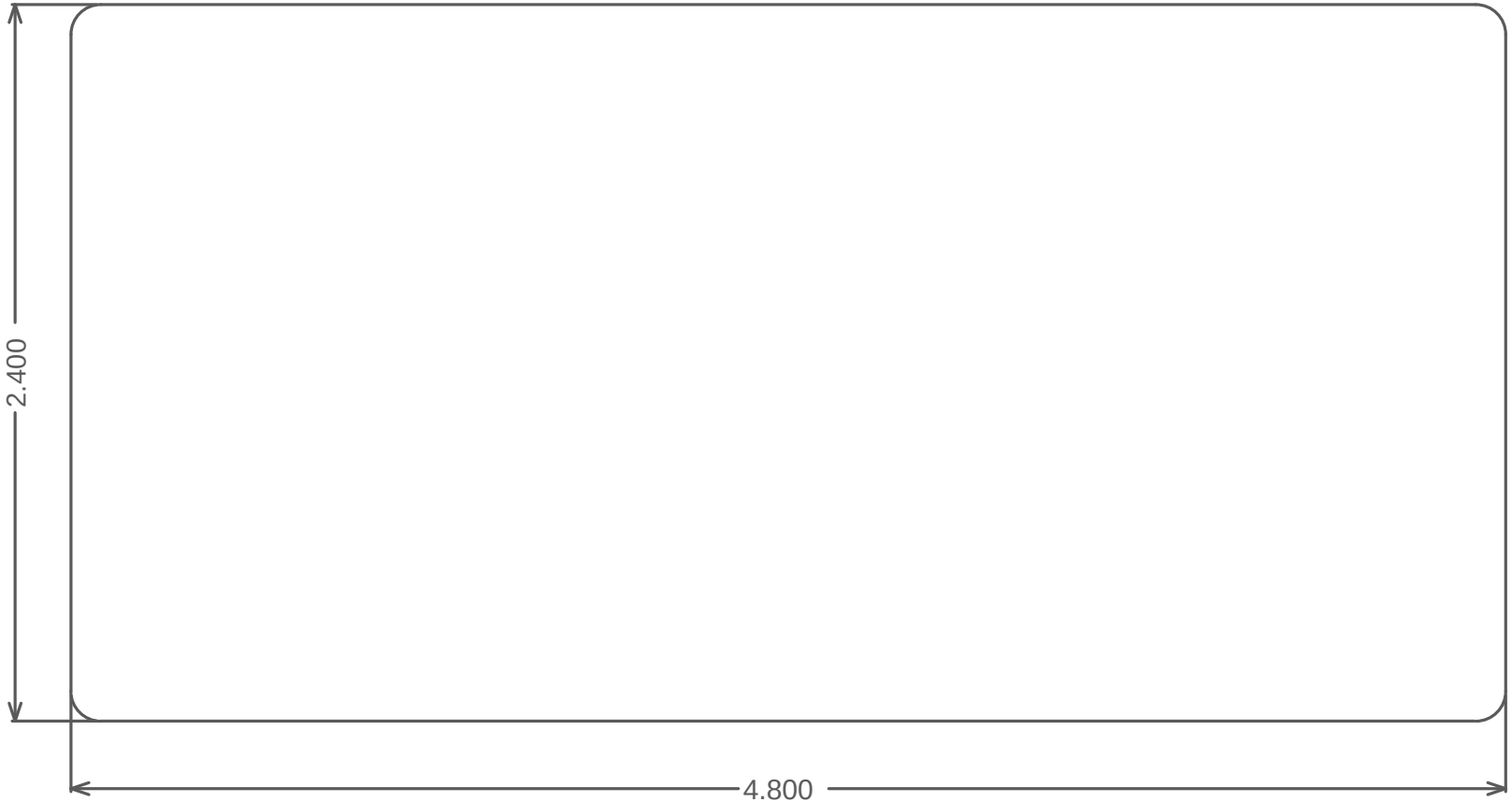
Drawn By: Jamus Griego		MICROCHIP
Engineer: John Mouton		
PartNumber: 10327	Project Title: *Curiosity LPC*	
Sheet Title: **		Designed with Altium Altium.com
Size: B	Sch #:03-10327	Date: 10/29/2020 7:30:32 PM
	Revision:2.2	Sheet 1 of 1
File: 03-10327 sheet 1.SchDoc		

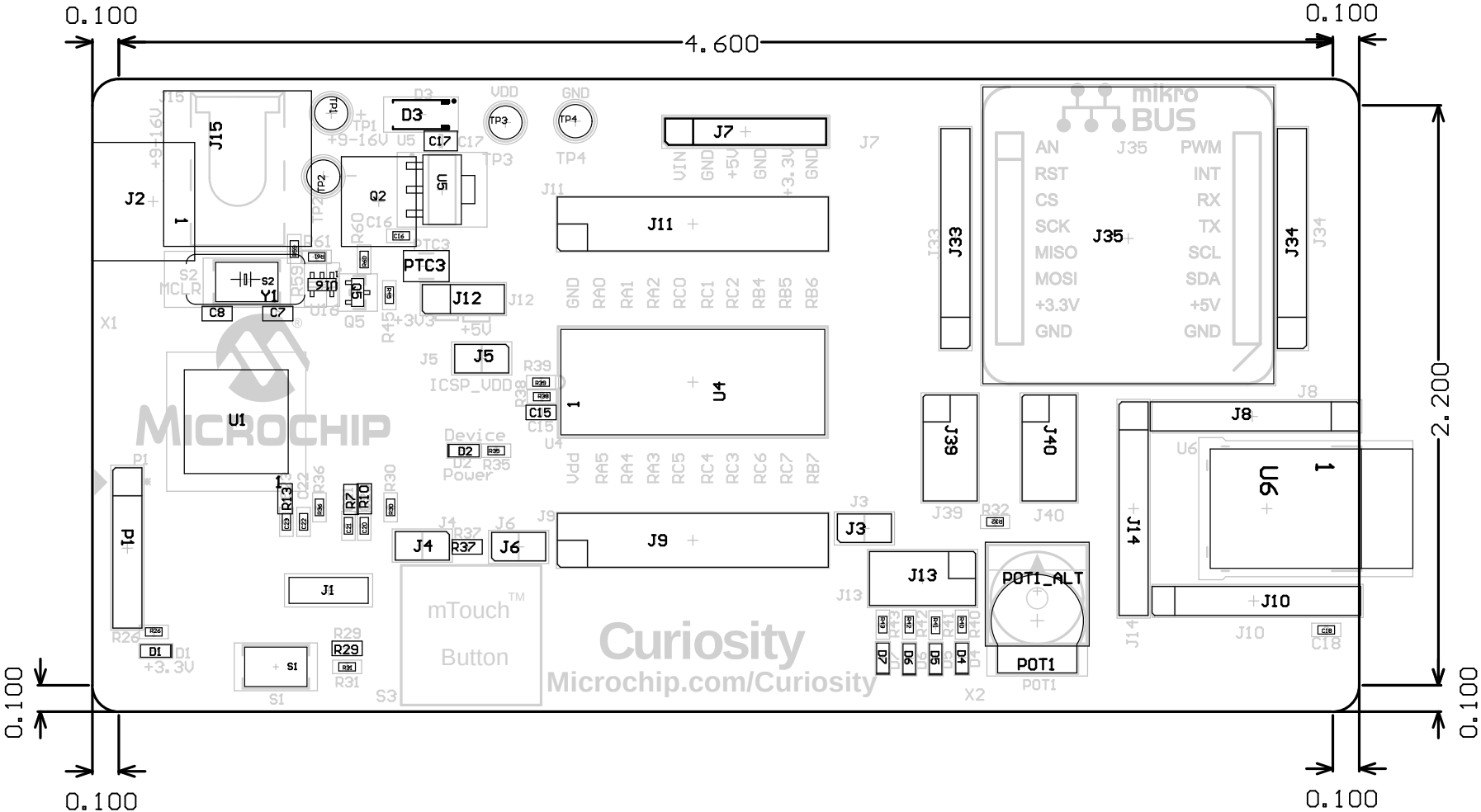
'PGED' = Programming/Emulation Data signal
'PGEC' = Programming/Emulation Clock signal

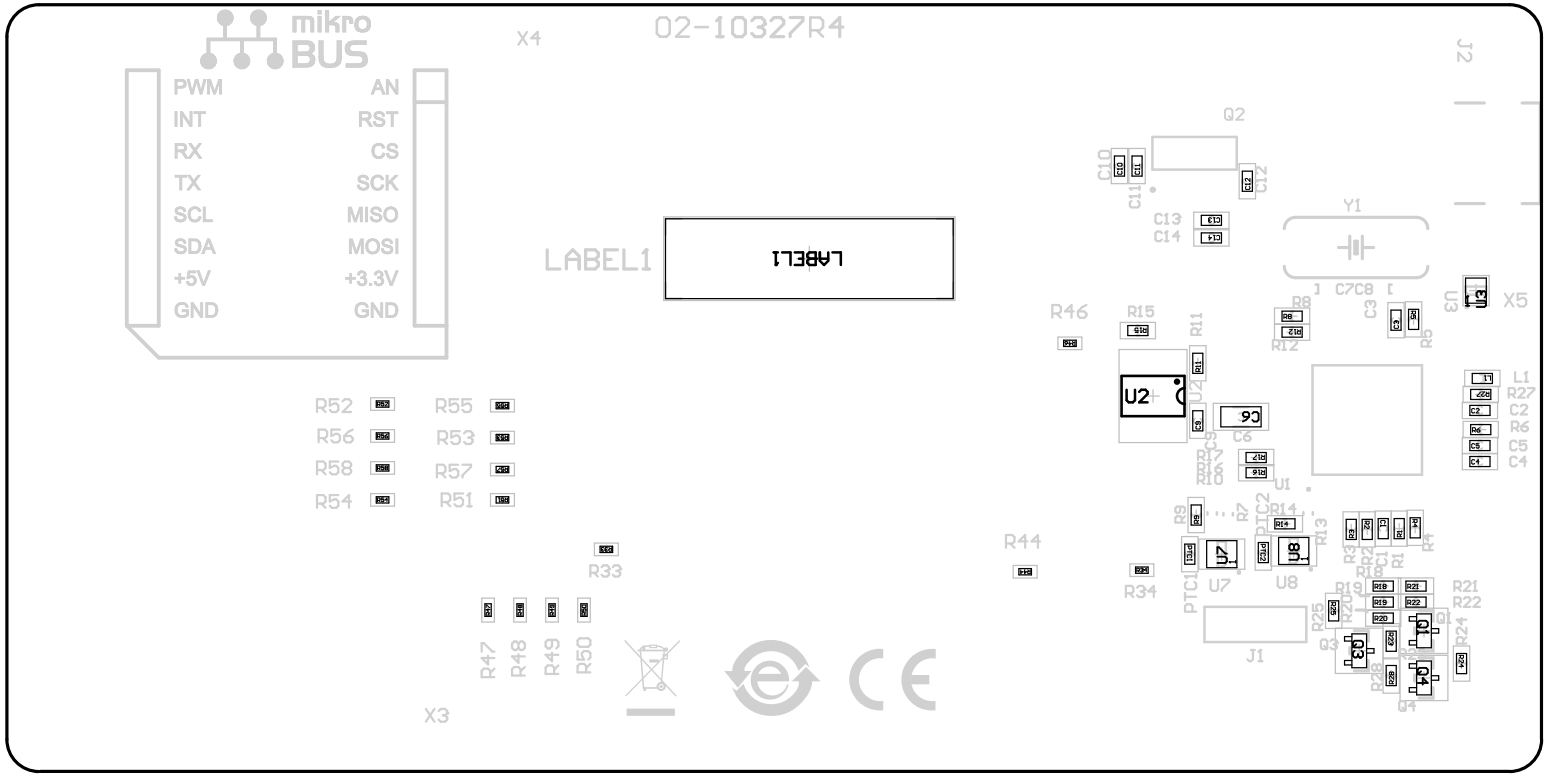


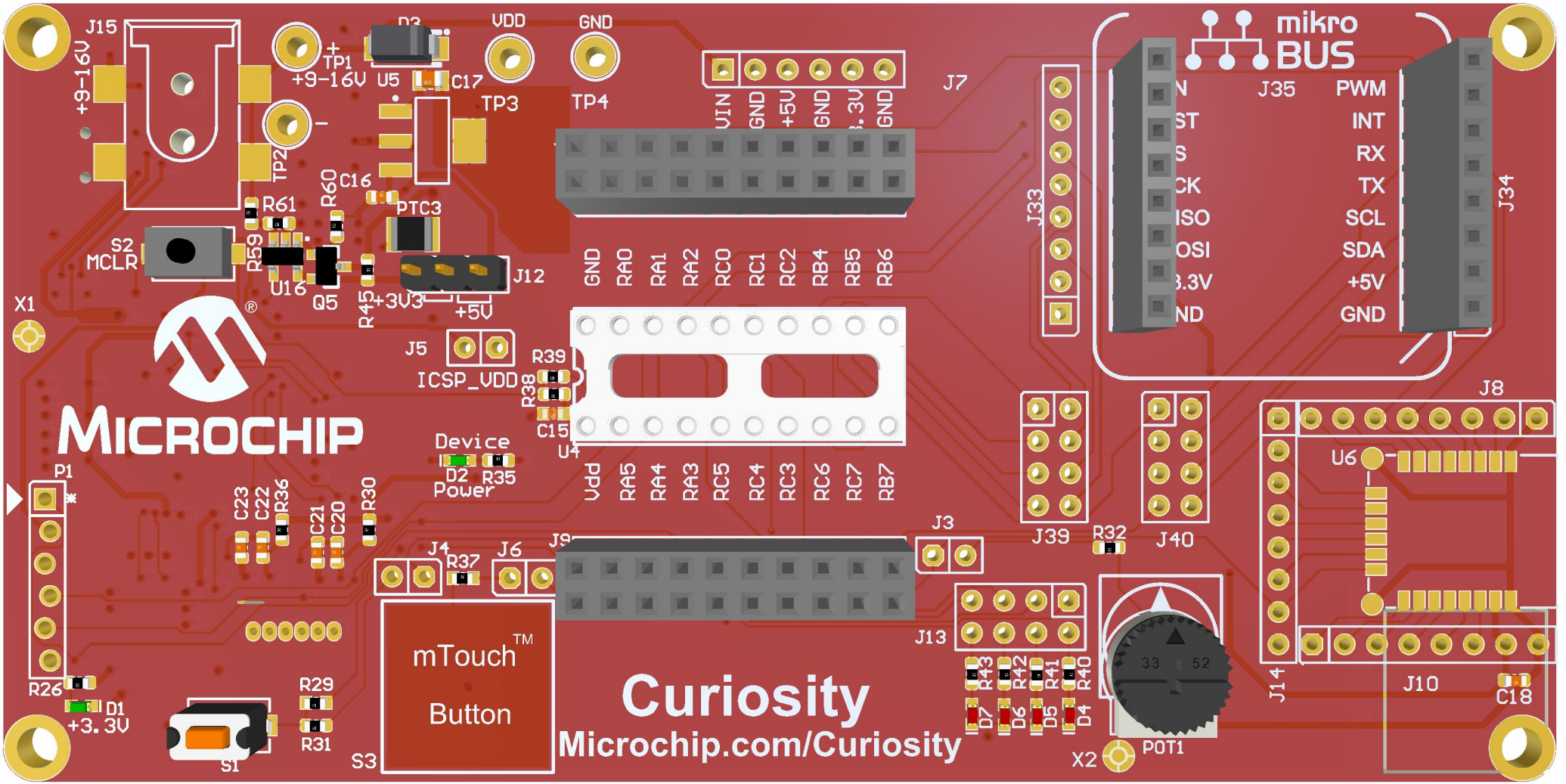
LABEL1
02- -R1
JIT
LABEL Need Help Small

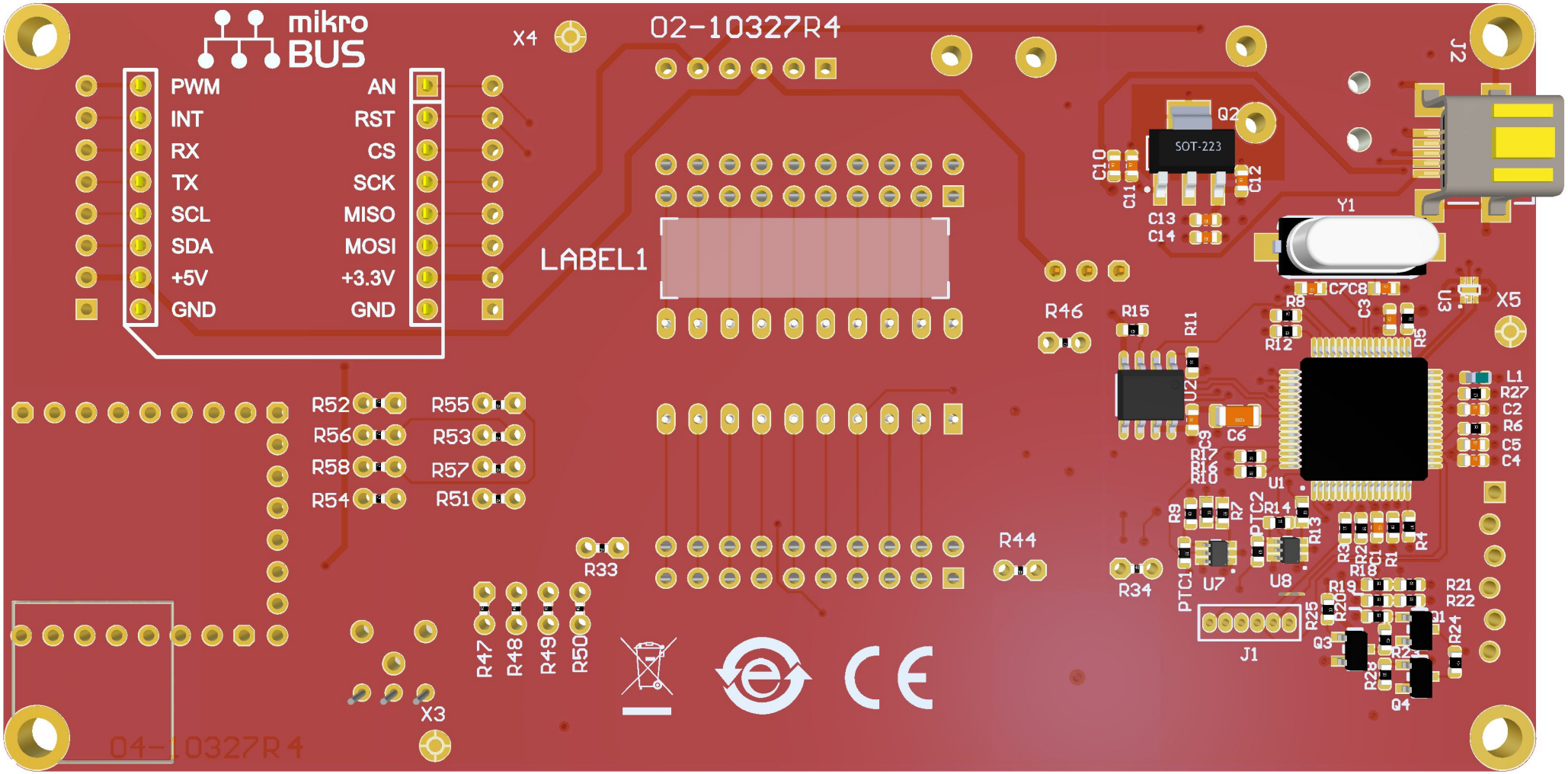
Drawn By: Jamus Griego	MICROCHIP
Engineer: John Mouton	
PartNumber: 10327	Project Title: *Curiosity LPC*
Sheet Title: **	Designed with Altium — Altium.com
Size: B	Sch #:03-10327 · Date: 10/29/2020 7:30:32 PM · Revision: 2.2 · Sheet 2 of 2
File: 03-10327 Sheet 2.SchDoc	

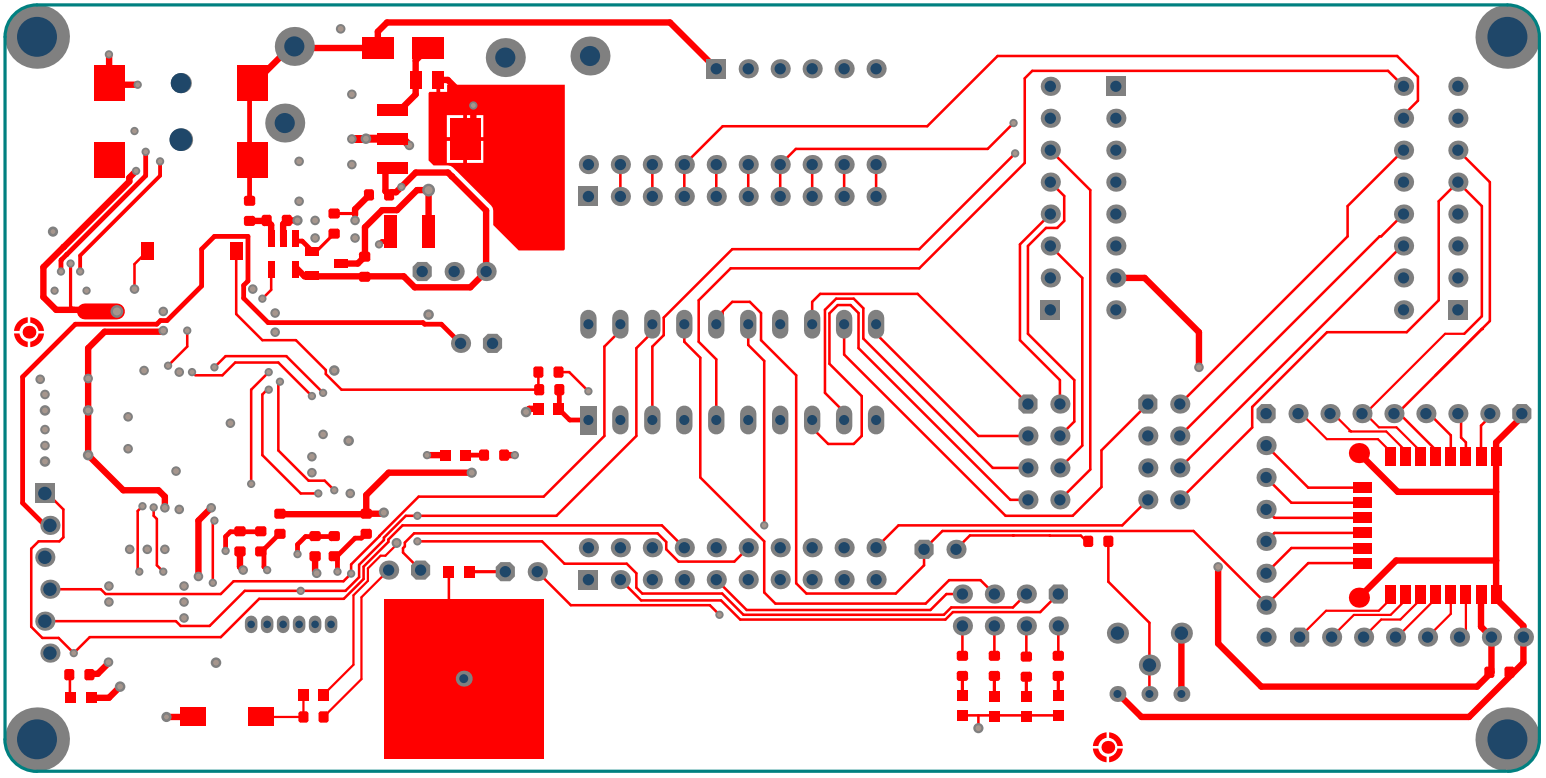

2.400
4.800

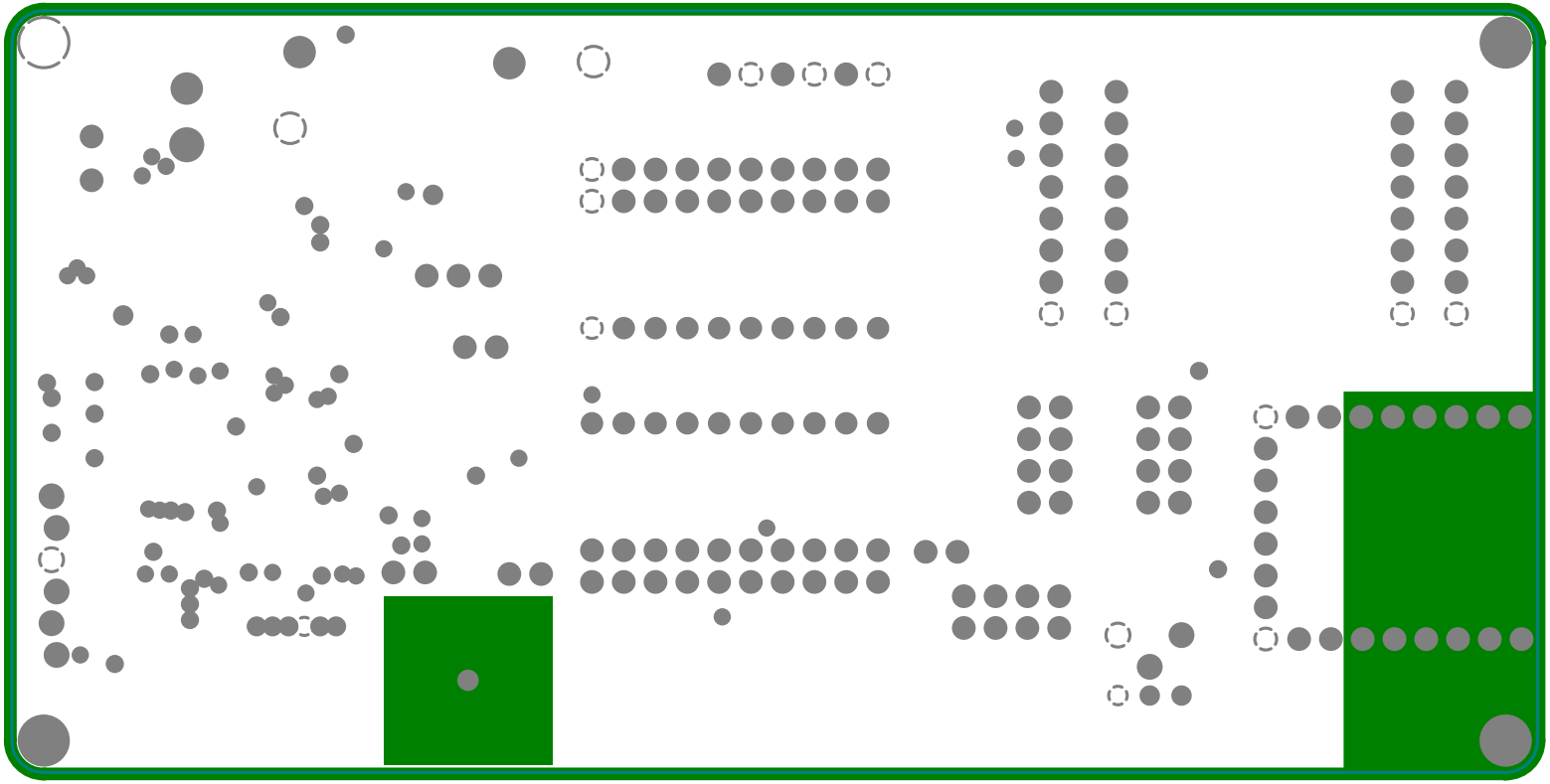

0.100
4.600
0.100
2.200
0.100
0.100
0.100
0.100
mikro BUS
AN
RST
CS
SCK
MISO
MOSI
+3.3V
GND
PWM
INT
RX
TX
SCL
SDA
+5V
GND
J35
J33
J34
J7
J11
J12
J5
J4
J6
J9
J3
J13
J39
J40
J8
J10
J14
U1
U4
U5
U6
Q2
Q5
Y1
PTC3
POT1_ALT
POT1
mTouch™
Button
Curiosity
Microchip.com/Curiosity
MICROCHIP

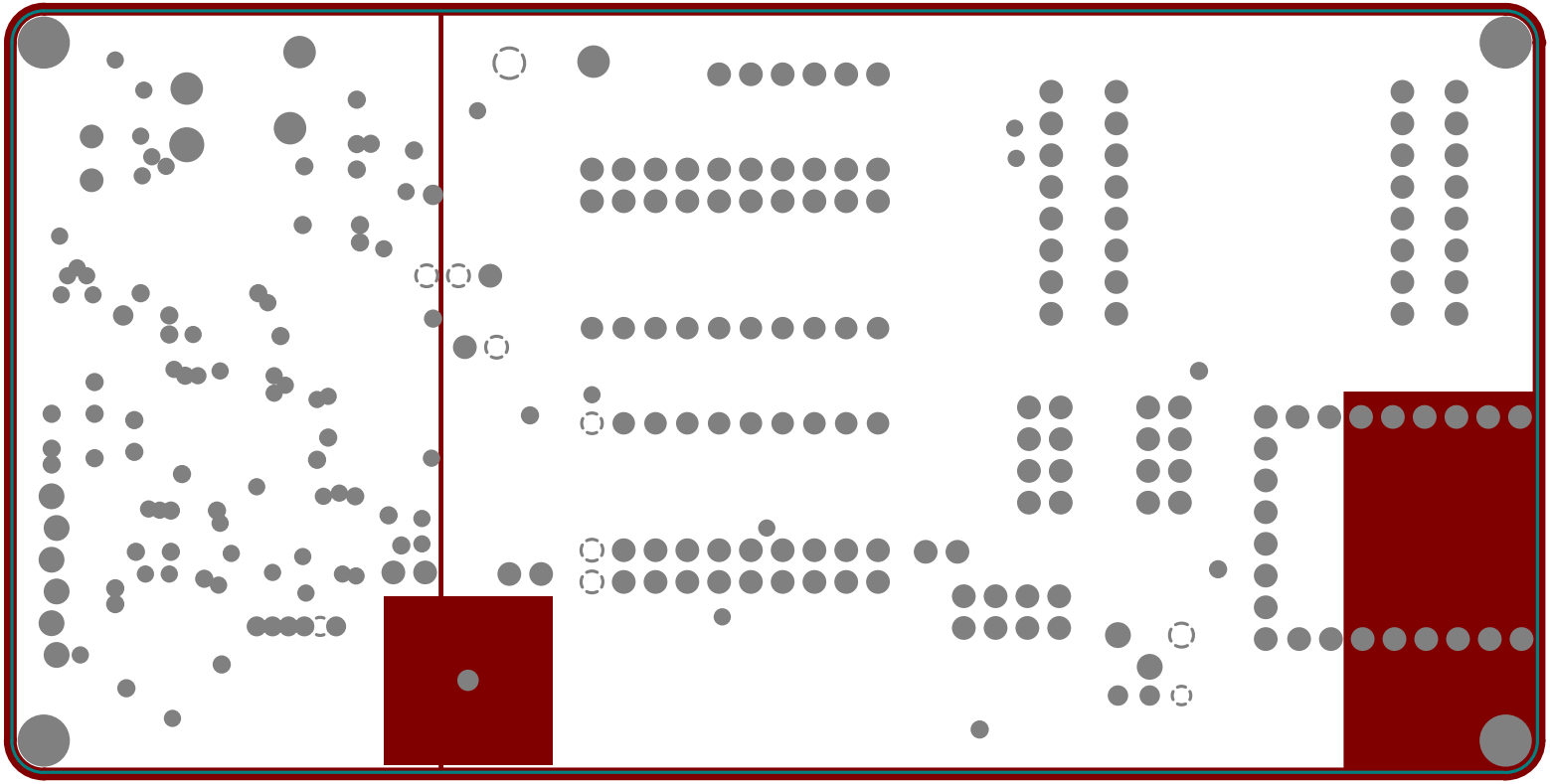

mikro BUS
PWM
INT
RX
TX
SCL
SDA
+5V
GND
AN
RST
CS
SCK
MISO
MOSI
+3.3V
GND
X4
02-10327R4
LABEL1
LABEL1
J2
Q2
Y1
X5
U3
L1
U2
C6
U1
U7
U8
J1
X3
R52
R55
R56
R53
R58
R57
R54
R51
R33
R47
R48
R49
R50
R46
R44
R34

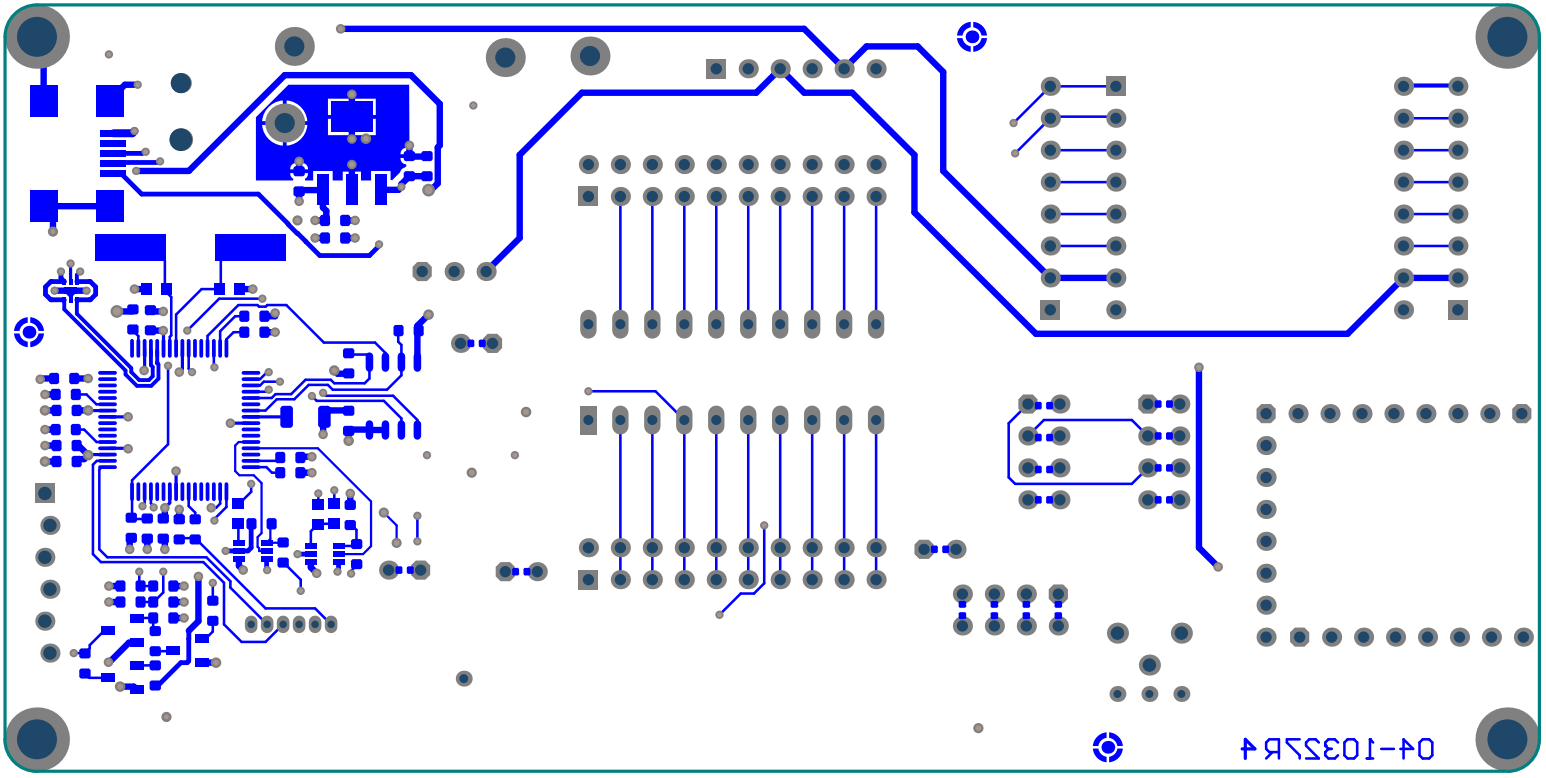

J15
+9-16V
TP1
+9-16V
D3
U5
C17
VDD
GND
TP3
TP4
J7
VIN
GND
+5V
GND
3.3V
GND
mikro
BUS
J35
PWM
INT
RX
TX
SCL
SDA
+5V
GND
J34
J33
TP2
R60
C16
R61
PTC3
R59
S2
MCLR
U16
Q5
R45
+3V3
J12
+5V
GND
RA0
RA1
RA2
RC0
RC1
RC2
RB4
RB5
RB6
X1
J5
ICSP_VDD
R39
R38
C15
U4
MICROCHIP
Device
D2
R35
Power
Vdd
RA5
RA4
RA3
RC5
RC4
RC3
RC6
RC7
RB7
J8
U6
P1
C23
C22
R36
C21
C20
R30
J4
R37
J6
J3
J39
R32
J40
J13
R43
R42
R41
R40
D7
D6
D5
D4
R26
D1
+3.3V
R29
R31
S1
S3
mTouch™
Button
Curiosity
Microchip.com/Curiosity
X2
POT1
J14
J10
C18


mikro BUS
PWM
INT
RX
TX
SCL
SDA
+5V
GND
AN
RST
CS
SCK
MISO
MOSI
+3.3V
GND
02-10327R4
04-10327R4
LABEL1
X4
X3
X5
J2
J1
Y1
Q2
SOT-223
Q3
Q4
U1
U2
U3
U7
U8
PTC1
PTC2
C10
C11
C12
C13
C14
C7C8
C3
C6
R52
R56
R58
R54
R55
R53
R57
R51
R33
R34
R44
R46
R15
R11
R8
R12
R5
R47
R48
R49
R50








04-10327R4

Component list

Bill of Materials Fitted for Project [02-10327 Curiosity.PrjPcb] (No PCB Document Selected)

Source Data From: 02-10327 Curiosity.PrjPcb
Project: 02-10327 Curiosity.PrjPcb
Variant: None

Report Date: 10/29/2020 7:30 PM
Print Date: 10/29/2020 7:28:41 PM

Designator	Quantity	Value	Manufacturer 1	Manufacturer Part Number 1	Description
@U4	1	PIC16F18346-I/P	Microchip Technology	PIC16F18346-I/P	
C1, C2, C3, C4, C5, C13, C15, C16, C20, C21, C22, C23	12	0.1uF	KEMET	C0603C104M3RACTU	Capacitor
C6	1	10uF	TDK	C3216X5R1C106K	Capacitor
C7, C8	2	22pF	Murata	GRM1885C2A220JA01D	Capacitor
C9, C11, C12	3	1uF	Taiyo Yuden	EMK107B7105KA-T	Capacitor
C10	1	0.22uF	KEMET	C0603C224Z4VACTU	Capacitor
C14	1	0.01uF	Samsung	CL10B103KO8NNNC	Capacitor
C17	1	0.22uF	Murata	GCM21BR71H224KA37L	Capacitor
C18	1	4.7uF	Taiyo Yuden	LMK107BJ475KA-T	Capacitor
D1, D2	2	GREEN	Lite-On	LTST-C190GKT	LED
D3	1	US1M	Vishay	US1M-E3/61T	
D4, D5, D6, D7	4	RED	Lite-On	LTST-C190KRKT	LED
J1	1	HDR-1.27 Male 1x6	Sullins	GRPB061VWVN-RC	
J2	1	USB MINI-B Female	Hirose	UX60-MB-5ST	Connector USB
J3, J4, J5, J6	4	HDR-2.54 Male 1x2	Molex Inc	22284020	
J7	1	HDR-2.54 Female 1x6	FCI	75915-306LF	
J8, J10, J14	3	HDR-2.54 Male 1x8	FCI	68001-108HLF	
J9, J11	2	HDR-2.54 Female 2x10	Samtec, SHENZHEN JUNHAO ENGINEERING	SSW-110-01-S-D, 214-2010-0021-000	
J12	1	HDR-2.54 Male 1x3	JAMECO VALUEPRO	7000-1X3SG-R	
J13, J39, J40	3	HDR-2.54 Male 2x4	Samtec	TSW-104-08-L-D	
J15	1	PJ-002BH-SMT	CUI Inc	PJ-002BH-SMT-TR	
J33, J34	2	HDR-2.54 Female 1x8	Sullins	PPTC081LFBN-RC	CON HDR-2.54 Female 1x8 Gold TH
J35	1	mikroBUS	Sullins Connector Solutions	PPTC081LFBN-RC	
L1	1	220 OHM	Murata Electronics North America	BLM18AG221SN1D	
LABEL1	1	LABEL Need Help Small			
P1	1	HDR-2.54 Male 1X6 STAGGERD	Sullins Connector Solutions	PBC06SAAN	
POT1	1	10k	Bourns	3352T-1-103	Resistor Var
POT1_ALT	1	10K	Bourns Inc.	3386P-1-103TLF	Resistor Var
PTC1, PTC2	2	50mA	Bel Fuse	0ZCM0005FF2G	
PTC3	1	500mA	Bel Fuse Inc	0ZCB0050FF2G	
Q1	1	MMBT3906	Central Semiconductor	CMPT3906-PST-LEADFREE	Transistor Bipolar
Q2	1	MCP1703/3.3V	Microchip	MCP1703-3302E/DB	LDO
Q3, Q4	2	MMBT3904	Diodes Incorporated	MMBT3904-7	Transistor Bipolar
Q5	1	FDN340P	Fairchild Semiconductor	FDN340P	Transistor MOS
R1, R26, R40, R41, R42, R43	6	1k	Panasonic	ERJ-3EKF1001V	Resistor
R2, R6, R8, R11, R12, R19, R23, R25, R27, R28, R29, R37, R38, R60, R61	15	10k	NIC Components	NRC06F1002TRF	Resistor
R3, R31, R32, R35, R39	5	470R	Yageo	RC0603FR-07470RL	Resistor
R4	1	3.16k	Panasonic	ERJ-3EKF3161V	Resistor
R5, R24, R30, R36	4	100R	Panasonic	ERJ-3EKF1000V	Resistor
R7, R10, R13	3	330R	ROHM	MCR03EZPFX3300	Resistor
R9, R14	2	4.7k	Panasonic	ERJ-3EKF4701V	Resistor
R15	1	2.2k	ROHM	MCR03EZPFX2201	Resistor
R16, R20	2	100k	Panasonic	ERJ-3EKF1003V	Resistor
R17	1	200k	Panasonic	ERJ-3EKF2003V	Resistor
R18	1	3.92k	Panasonic	ERJ-3EKF3921V	Resistor
R21, R22	2	2.21k	Panasonic	ERJ-3EKF2211V	Resistor
R33, R34, R44, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58	16	0R	Panasonic	ERJ-2GE0R00X	Resistor
R45	1	0R	NIC Components	NRC06Z0TRF	Resistor
R59	1	2k	Panasonic	ERJ-3EKF2001V	Resistor
S1, S2	2	TACT SPST	C&K Components	KSR231GLFS	Switch
SCR1, SCR2, SCR3, SCR4	4	Phillips Screw 1/4"	Keystone Electronics	9900	
STANDOFF1, STANDOFF2, STANDOFF3, STANDOFF4	4	1/4" Standoff Nylon	Keystone Electronics	1902A	
TP1, TP3	2	TP LOOP Red	Keystone	5010	Test Point
TP2, TP4	2	TP LOOP Black	Keystone	5011	Test Point
U1	1	PIC24FJ256GB106	Microchip	PIC24FJ256GB106T-I/PT	PIC24F
U2	1	25LC256	Microchip	25LC256T-E/SN	EEPROM SPI
U3	1	VBUS054B-HS3-GS08	Vishay Semiconductor Diodes Division	VBUS054B-HS3-GS08	
U4	1	110-91-320-41-001	Mill-Max	110-99-320-41-001	Connector
U5	1	LM340MP-5.0	National Semiconductor	LM340MP-5.0	LDO
U6	1	RN4020	Microchip Technology	RN4020-V/RM	
U7, U8	2	SN74LVC1T45DCKR	Texas Instruments	SN74LVC1T45DCKR	
U16	1	MCP6561	Microchip	MCP6561T-E/OT	OPAMP
Y1	1	12MHz	Abracon	ABLS-12.000MHZ-B2-T	Crystal
	148				

Notes